MILITARY SPECIFICATION

SCREW, COLLAR, 12-POINT, EXTERNAL WRENCHING

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. - This specification covers external wrenching 12-point collar screws.

1.2 Classification. -

1.2.1 External wrenching 12-point collar screws shall be the types and classes specified in the applicable specification sheet (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35 - Passivation Treatment for Austenitic, Ferritic, and Martensitic Corrosion Resisting Steel (Fastening Devices).
QQ-P-416 - Plating, Cadmium (Electrodeposited).

MILITARY

MIL-H-3982 - Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No Go).

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability.
FED-STD-151 - Metals; Test Methods.

FSC 5305

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MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definition.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS HANDBOOK

H-28, Part I - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

AMERICAN STANDARDS ASSOCIATION, INC.

ASA B46.1 (1962) - Surface Texture (Surface Roughness, Waviness and Lay).

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. - The materials used shall be such as to produce 12-point collar screws in compliance with the requirements of this specification and as specified in the applicable specification sheet. External wrenching 12-point collar screws shall be made of alloy steel or corrosion resistant steel in accordance with FED-STD-66 (see 6.2).

3.1.1 Alloy steel. - Alloy steel external wrenching 12-point collar screws shall be of a chemical composition which can be heat treated to meet the mechanical properties specified in 3.2.1.

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3.1.2 Corrosion resistant steel. - Corrosion resistant steel external wrenching 12-point collar screws shall be manufactured from a steel composition such as type 304, 305, or 316 as specified in FED-STD-66,or a composition 16:18 or 18:8 chromium-nickel alloy steel (developed for cold-heading), i.e., austenitic steel which meets the mechanical properties specified in 3.2.2.

3.2 Mechanical properties. -

3.2.1 .Alloy steel. - Mechanical properties of alloy steel external wrenching 12-point collar screws shall conform to the following:

(a) Ultimate tensile strength.....160,000 p.s.i. min.
(b) Yield strength (0.2 percent offset).....130,000 p.s.i. min.
(c) Hardness, Rockwell (after heat treatment).36 to 43 Rockwell "C".
(d) Elongation.....12 percent min.

3.2.2 Corrosion resistant steel. - Unless otherwise specified (see 6.2), mechanical properties of corrosion resistant steel, external wrenching 12-point collar screws shall conform to the following for diameters through 5/8 inch:

(a) Ultimate tensile strength.....80,000 p.s.i. min.
(b) Yield strength (0.2 percent offset).....30,000 p.s.i. min.
(c) Hardness, Rockwell.....B80, min.
(d) Elongation.....30 percent min.

For diameters over 5/8 inch, ultimate tensile strength shall be 70,000 p.s.i. minimum and have a hardness of Rockwell B74 minimum.

3.2.3 Wedge tensile strength. - Twelve point collar screws made of alloy steel or corrosion resistant steel shall conform to the tensile strength requirements of 3.2.1 and 3.2.2 as indicated by samples tested in accordance with 4.4.3.1.

3.3 Protective coatings. - External wrenching 12-point collar screws shall be furnished uncoated (black heat treated finish), passivated, or coated as specified in the applicable specification sheet.

3.3.1 Passivation. - Corrosion resistant steel external wrenching 12-point collar screws shall be passivated in accordance with QQ-P-35.

3.3.2 Cadmium plating. - When specified (see 6.2), external wrenching 12-point collar screws shall be cadmium plated. Cadmium plate shall conform to type II, class 3 of QQ-P-416.

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3.3.3 Phosphate coating. - When specified (see 6.2), external wrenching 12-point collar screws shall be phosphate coated. Phosphate coatings shall conform to type Z, class 2 of MIL-P-16232.

3.3.4 Hydrogen embrittlement. - Alloy steel external wrenching 12-point collar screws required to be electroplated or phosphate coated shall be subjected to a relief treatment in accordance with the applicable coating or plating specification.

3.4 Surface roughness. - The surface roughness of the flanks and roots of thread shall be 63 micro-inches RHR or less, in accordance with ASA B46.1. The surface roughness of external wrenching 12-point collar screws, before passivating or coating shall be 125 micro-inches or less on the shank, fillets and head bearing surfaces.

3.5 Magnetic permeability. - The magnetic permeability shall be as specified in the applicable specification sheet.

3.6 Dimensions and tolerances. -

3.6.1 Limits of size. - External wrenching 12-point collar screws shall conform to the dimensions specified in the applicable specification sheet.

3.6.2 Screw threads. - The screw thread series and class shall be in accordance with H28 and as specified in the applicable specification sheet.

3.7 Metallographic requirements. -

3.7.1 Decarburization. - Unless otherwise specified (see 6.2), the partial decarburization of threads of alloy steel collar screws shall not exceed the amount shown on figure 1.

3.8 Point chamfer. - The point shall be flat and chamfered from approximately 1/64 inch below the minor diameter of the thread. The flat shall be normal to the axis of the screw. The edge between the flat and chamfer may be slightly rounded. The length of the chamfer shall be from 1/2 to 1-1/2 threads and may be distorted slightly from thread forming.

3.9 Bearing surface. - The plane of the bearing surface shall be perpendicular to the axis of the screw within a maximum deviation of one degree.

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3.10 Concentricity. -

3.10.1 The collar shall be concentric with the body within 1 percent of the maximum screw diameter, (2 percent total runout) or 0.010 inch total runout, whichever is greater (see 4.4.9).

3.11 Defects. -

3.11.1 Cracks. - External wrenching 12-point collar screws having visible circumferential cracks in the shank, fillet or thread root area shall not be acceptable. A crack shall be defined as a crystalline break through the grain boundary without the inclusion of foreign elements.

3.11.2 Seams. - External wrenching 12-point collar screws having seams, except in locations specified hereinafter, which exceed the tolerance limits specified below shall not be acceptable. Seams up to the depth indicated, which have been rolled beneath the root of the thread shall be acceptable. Seams or laps as listed in the following tabulation are acceptable in the crest portion of the threads.

Limits of seam depth

Size (inch)	*Depth (inch)*
5/16 and less	0.010
3/8	.012
7/16	.014
1/2	.016
5/8 and over	.004*

* For each 1/8 inch of diameter.

3.11.3 Screw heads. - External wrenching 12-point collar screws having seams, inclusions, nicks or gouges along the top or sides of the head within twice the limits specified in 3.11.2 for depth are acceptable. Screws with seams in the bearing surface exceeding the limits of 3.11.2 are not acceptable. Any mutilation of the bearing surface sufficient to prevent seating of the screw shall not be acceptable.

3.11.4 Head and shank junction. - Only longitudinal seams of the depths specified in 3.11.2 shall be acceptable. Tool marks or undercuts of depths which do not exceed the following are acceptable provided that they blend out into the shank with no gouges or corners which will prevent proper ins installation of the screw.

Limits of tool marks or undercuts

Size (inch)	*Depth (inch)*
Up to 3/8 inclusive	0.003
Over 3/8 to 5/8	.004
Over 5/8 to 7/8	.005
Over 7/8	.006

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification: where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms. - Inspection terms and definitions shall be in accordance with MIL-STD-109.

4.2 Sampling. -

4.2.1 Inspection lot. - All external wrenching 12-point collar screws of the same type, material, protective coating, thread series, and size produced under essentially the same conditions, shall be considered a lot for the purposes of inspection.

4.2.2 Sampling for examination of screws. - A random sample of external wrenching 12-point collar screws shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-1. The acceptable quality level (AQL) shall be as specified in table I.

Table I - Classification of defects in accordance with MIL-STD-105.

Categories	Defects	Method of inspection
Critical	AQL = 1.0 percent	
1	Circumferential cracks (3.11.1)	Metallographic microscope, magnetic particle or penetrant.
2	Seams (3.11.2)	Metallographic microscope, magnetic particle or penetrant.

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Table I - Classification of defects in accordance with MIL-STD-105 (cont'd.)

Categories	Defects	Method of inspection
Major	AQL = 2.5 percent	
101	Non-conformance to thread requirements (3.6.2).	Gage
102	Non-conformance to head dimensions for collar diameter, across corners, or across flats (3.6.1).	Gage
Minor	AQL = 4.0 percent	
201	Non-conformance to thread length (3.6.1).	SIE 1/
202	Non-conformance to body length (3.6.1).	SIE
203	Non-conformance to screw length (3.6.1).	SIE
204	Non-conformance to head height (3.6.1).	SIE
205	Non-conformance to collar thickness (3.6.1).	SIE
206	Non-conformance to concentricity requirements (3.6.1)	SIE
207	Non-conformance to chamfers or fillet requirements (3.6.1).	SIE
208	Non-conformance to surface roughness limitations (3.4).	Visual
209	Protective coating missing	2/

1/ Standard inspection equipment.
2/ See 4.4.6 for tests of protective coatings.

4.2.3 Sampling for mechanical properties and metallographic tests. - A random sample of external wrenching 12-point collar screws shall be taken from each lot in accordance with MIL-STD-105 at inspection level S-1. The AQL shall be 1.5 percent defective.

4.2.4 Sampling for protective coating tests. - Sampling for tests of protective coatings shall be in accordance with the applicable specification as specified in 3.3.1 through 3.3.3.

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4.2.5 Sampling for examination of packaging and packing. - Sampling for examination tests of preservation, packaging, packing, and marking for shipment shall be in accordance with MIL-H-3982.

4.2.6 Chemical analysis. - Each sample selected in accordance with 3.1.1 and 3.1.2 shall be analyzed by the wet chemical or spectrographic methods in accordance with method 111 or 112, respectively, of FED-STD-151. In case of dispute, chemical analysis by wet chemical methods shall be the basis for acceptance.

4.3 Examination. -

4.3.1 Visual and dimensional inspections. - Each external wrenching 12-point collar screw taken as specified in 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with table I. Any screw in the sample containing one or more defects shall be rejected and if the number of defective screws in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.3.2 Packaging and packing. - Examination and tests of preservation, packaging, and packing shall be in accordance with MIL-H-3982.

4.4 Test procedures. -

4.4.1 The tests specified in 4.4.2.1, 4.4.3, and 4.4.3.1 shall be performed by the screw supplier.

4.4.2 Mechanical property test. - Unless otherwise specified (see 6.2), the hardness test of 4.4.2.1 may be performed in lieu of those in 4.4.3 and 4.4.3.1. If a lot has failed to meet the hardness test of 4.4.2.1 such additional tests as are specified in 4.4.3 and 4.4.3.1 shall be used to determine compliance with mechanical properties.

4.4.2.1 Hardness test. - Hardness test for external wrenching 12-point collar screws shall be conducted in accordance with FED-STD-151, on a section through the threaded portion of the screw, taken one nominal diameter from the point. Hardness shall be measured at a point on the section, one quarter diameter from the axis of the screw. For screws three quarter inch and larger, test may be made on a test specimen.

4.4.3 Tensile test. - Except as otherwise specified (see 6.2), the tensile test shall be conducted in accordance with method 211.1 of FED-STD-151.

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4.4.3.1 Wedge tensile strength test. - Sample external wrenching 12-point collar screws shall be placed in the tension test fixture, (see figure 2), a wedge (see 4.4.3.2), placed under the bolt head and tensile test loading applied continuously until failure. Failure of the screws in the fillet area, or fracture in the collar or head shall be cause for rejection. A screw shall be considered as passing the test when the following conditions are met:

(a) Failure does not occur until the minimum breaking loads specified in tables II, III, IV, and V are reached.
(b) Failure occurs in the threads or shank.

Table II - Minimum breaking loads of alloy steel screws, coarse thread series.

Nominal size	Threads per inch	Mean stress area 1/ of thread (see H28)	Ultimate load capacity minimum (see 3.2.1) 2/ and 4/	Yield load capacity minimum (see 3.2.1) 3/ and 4/
Inches		Inches	Pounds	Pounds
1/4	20 UNC	.03093	4,950	4,020
5/16	18 UNC	.05121	8,190	6,660
3/8	16 UNC	.07587	12,100	9,860
7/16	14 UNC	.10429	16,700	13,500
1/2	13 UNC	.13942	22,300	18,100
9/16	12 UNC	.17903	28,600	23,300
5/8	11 UNC	.22260	35,600	28,900
3/4	10 UNC	.32991	52,800	42,900
7/8	9 UNC	.45606	73,000	59,300
1	8 UNC	.59871	95,800	77,800
1-1/8	7 UNC	.75491	120,800	98,100
1-1/4	7 UNC	.95951	153,500	124,700
1-3/8	6 UNC	1.14341	183,000	148,600
1-1/2	6 UNC	1.39239	222,800	181,000

1/ $A_s = 3.1416 \left(\frac{E\ MIN}{2} - \frac{3H}{16}\right)^2$

2/ Based on ultimate tensile strength minimum.
3/ Based on yield strength minimum.
4/ Based on static loading.

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Table III - Minimum breaking loads of alloy steel screws, fine thread series.

Nominal size	Threads per inch	Mean stress area 1/ of thread (see H28)	Ultimate load capacity minimum (see 3.2.1) 3/ and 5/	Yield load capacity minimum (see 3.2.1) 4/ and 5/
Inches		Inches	Pounds	Pounds
1/4	28 UNF	.03553	5,680	4,620
5/16	24 UNF	.05690	9,100	7,400
3/8	24 UNF	.08629	13,800	11,200
7/16	20 UNF	.11682	18,700	15,200
1/2	20 UNF	.15768	25,200	20,500
9/16	18 UNF	.20026	32,000	26,000
5/8	18 UNF	.25282	40,400	32,900
3/4	16 UNF	.36885	59,000	48,000
7/8	14 UNF	.50429	80,700	65,500
1	14 NS 2/	.67404	107,800	87,600
1	12 UNF	.65675	105,100	85,400
1-1/8	12 UNF	.84841	135,700	110,300
1-1/4	12 UNF	1.06457	170,300	138,400
1-3/8	12 UNF	1.30524	208,800	169,700
1-1/2	12 UNF	1.57041	251,300	204,100

1/ $A_S = 3.1416 \left(\frac{E\ MIN}{2} - \frac{3H}{16}\right)^2$

2/ The 1-14 thread is a national special thread and should not be used for new design.

3/ Based on ultimate tensile strength minimum.

4/ Based on yield strength minimum.

5/ Based on static loading.

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Table IV - Minimum breaking loads of class 300 series corrosion resistant steel screws, coarse thread series.

Nominal size	Threads per inch	Mean stress area 1/ of thread (see H28)	Ultimate load capacity minimum (see 3.2.2) 2/ and 4/	Yield load capacity minimum (see 3.2.2) 3/ and 4/
Inches		Inches	Pounds	Pounds
1/4	20 UNC	.03180	2,540	950
5/16	18 UNC	.05240	4,190	1,570
3/8	16 UNC	.07750	6,200	2,320
7/16	14 UNC	.10630	8,500	3,190
1/2	13 UNC	.14190	11,300	4,260
9/16	12 UNC	.18200	14,500	5,460
5/8	11 UNC	.22600	18,100	6,780
3/4	10 UNC	.33400	23,400	10,000
7/8	9 UNC	.46200	32,300	13,800
1	8 UNC	.60600	42,400	18,200
1-1/8	7 UNC	.76300	53,400	22,900
1-1/4	7 UNC	.96900	67,800	29,100
1-3/8	6 UNC	1.15500	80,800	34,600
1-1/2	6 UNC	1.40500	98,300	42,100

1/ $A_s = 3.1416 \left(\frac{E}{2} - \frac{3H}{16}\right)^2$

2/ Based on ultimate tensile strength minimum.
3/ Based on yield strength minimum.
4/ Based on static loading.

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Table V - Minimum breaking loads of class 300 series corrosion resistant steel screws, fine thread series.

Nominal size	Threads per inch	Mean stress area 1/ of thread (see H28)	Ultimate load capacity minimum (see 3.2.2) 3/ and 5/	Yield load capacity minimum (see 3.2.2) 4/ and 5/
Inches		Inches	Pounds	Pounds
1/4	28 UNF	.03640	2,910	1,090
5/16	24 UNF	.05800	4,640	1,740
3/8	24 UNF	.08780	7,020	2,630
7/16	20 UNF	.11870	9,500	3,560
1/2	20 UNF	.15990	12,800	4,800
9/16	18 UNF	.20300	16,200	6,090
5/8	18 UNF	.25600	20,500	7,680
3/4	16 UNF	.37300	26,100	11,200
7/8	14 UNF	.50900	35,600	15,300
1	14 NS 2/	.67900	47,500	20,400
1	12 UNF	.66300	46,400	19,900
1-1/8	12 UNF	.85600	59,900	25,700
1-1/4	12 UNF	1.07300	75,100	32,200
1-3/8	12 UNF	1.31500	92,000	39,400
1-1/2	12 UNF	1.58100	110,700	47,400

1/ $A_s = 3.1416 \left(\frac{E}{2} - \frac{3H}{16}\right)^2$

2/ The 1-14 thread is a national special thread and should not be used for new design.

3/ Based on ultimate tensile strength minimum.

4/ Based on yield strength minimum.

5/ Based on static loading.

4.4.3.2 Wedge. - The wedge used in the wedge tensile test shall be as shown on figure 3.

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4.4.4 Test specimens. - External wrenching 12-point collar screws shall be tested full size preferably using an apparatus similar to that illustrated on figure 2.

4.4.4.1 Turned shank specimens. - For testing external wrenching 12-point collar screws having tensile properties greater than can be measured on available test equipment, the body shall be turned down around the axis of the screw to the gage dimension of the "R" type specimen (see 4.4.3) leaving the threads intact as illustrated on figure 4.

4.4.4.2 Type R1 specimens. - (See method 211.1 of FED-STD-151). For testing external wrenching 12-point collar screws too large for full size testing an "R" type test specimen of the largest size practicable shall be turned from the screw having its axis midway between the axis of the screw and the surface of the body as depicted by figure 5.

4.4.4.3 Ductility test specimens. - Test for elongation shall be conducted on a turned shank specimen (see figure 4) having dimensions equal to the largest practical "R" type specimen (see 4.4.3).

4.4.5 Metallographic test. - Unless otherwise specified (see 6.2), a metallographic test shall be made and conducted in accordance with 4.4.5.1.

4.4.5.1 Decarburization. - To test for decarburization, the threaded part shall be sectioned longitudinally through the axis and polished in such a manner that rounding of the thread edges is held to a minimum. The section shall be etched with the usual metallographic etchant (such as nital) and examined with a metallographic microscope. Measurements shall be made from the crest of the thread to the end of the decarburized zone. Such distance shall not exceed that indicated on figure 1. Measurements shall be made with a fixed scale micrometer eye piece. This does not apply to austenitic grades of stainless steel.

4.4.6 Protective coating test. - Test of protective coatings shall be conducted in accordance with the applicable specification specified in 3.3.1 through 3.3.3.

4.4.7 Hydrogen embrittlement. - The contractor shall furnish the Government certification that electroplated or phosphate coated alloy steel screws have been subjected to the hydrogen embrittlement relief treatment specified in 3.3.4.

4.4.8 Magnetic permeability. - The magnetic permeability of corrosion resistant steel external wrenching 12-point collar screws shall be determined by the use of an indicator in accordance with MIL-I-17214, or equivalent (see 3.5). This test shall be performed on the same samples selected for visual and dimensional inspection (see 4.3.1).

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4.4.9 Total runout test. - Total runout between thread, body and collar shall be checked by assembling screws into a compound gage as shown on figure 6.

4.4.10 Examination and test of preparation for delivery. - The preservation, packaging, packing and marking shall be examined or tested in accordance with MIL-H-3982.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - External wrenching 12-point collar screws shall be preserved and packaged in accordance with level A or C of MIL-H-3982, as specified (see 6.2).

5.2 Packing. - External wrenching 12-point collar screws packaged level A or C shall be packed level A, B, or C, as specified in MIL-H-3982 (see 6.2).

5.3 Marking. - In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. - Twelve point collar screws are intended for servicability where wrenching space is limited, where a 12-point wrenching surface facilitates torquing to clamp loads greater than normally required of 6-point and socket-drive cap screws, and "blind camming" contributes to rapid assembly and disassembly.

6.2 Ordering data. - Procurement documents should specify the following:

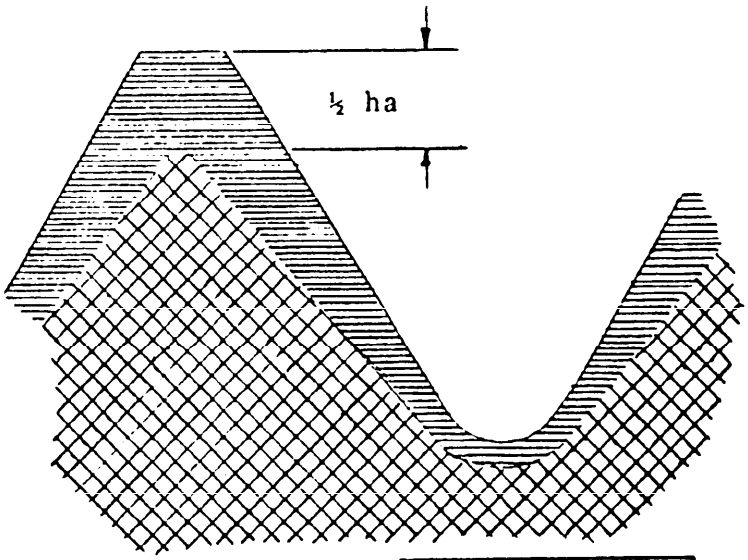
(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet.
(c) Type and class of collar screw required (see applicable specification sheet and 3.1).
(d) Mechanical properties of corrosion resistant steel, if other than specified (see 3.2.2).
(e) Protective coating required (see 3.3.2 and 3.3.3).
(f) If thread decarburization of alloy steel collar screws is other than specified (see 3.7.1).
(g) If mechanical property test is not required (see 4.4.2).

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(h) Whether tensile test is other than specified (see 4.4.3).
(i) If metallographic test is not required (see 4.4.5).
(j) Level of preservation, packaging and packing required see 5.1 and 5.2).
(k) Additional marking required (see 5.3).

Preparing activity:
Navy - SH
(Project 5305-N014Sh)

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Full Carbon Zone

Partial Decarburized Zone

Thds. per Inch	½ ha
28	.006
24	.007
20	.008
18	.009
16	.010
14	.012
13	.012
12	.014
11	.015
10	.016
9	.018
8	.020
7	.023
6	.027

½ ha = ½ Addendum.

Definition of addendum, figure 1. The addendum symbol "ha" is defined in Handbook H28, Screw Thread Standards for Federal Services as: "The distance measured perpendicularly to the axis, between the major and pitch diameter of cones, respectively."

Figure 1 - Decarburization classification of screw threads.

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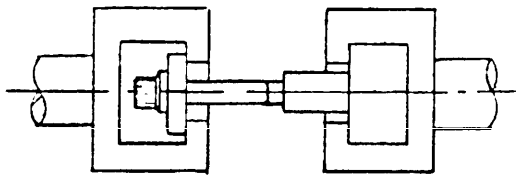


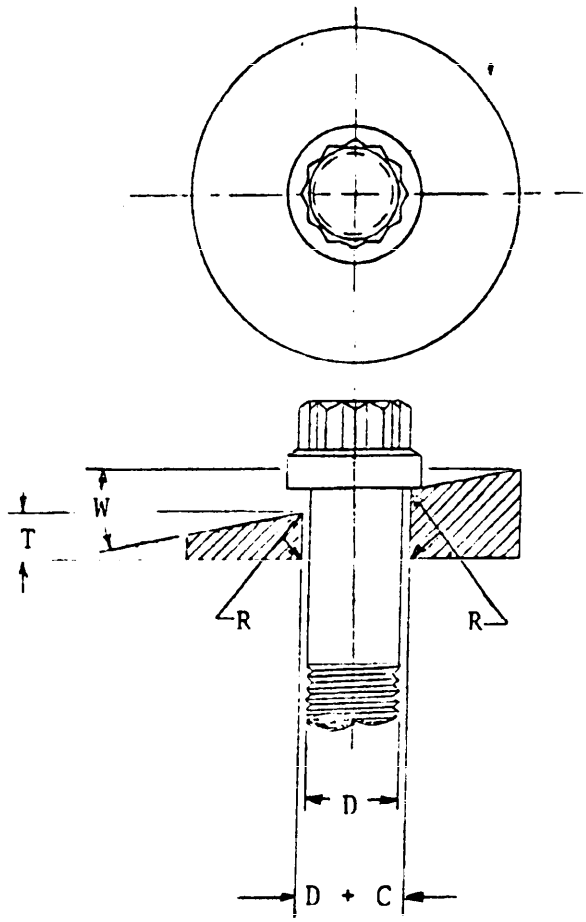
Figure 2 - Tension testing full size screw.

CLEARANCE (C) SHALL BE APPROX. AS FOLLOWS:

NOMINAL BOLT SIZE INCH	RADIUS INCH
1/4 TO 1/2	.030
9/16 TO 3/4	.050
7/9 TO 1-1/4	.063
1-3/8 TO 1-1/2	.094

MAXIMUM RADIUS R, SHALL BE AS FOLLOWS:

NOMINAL BOLT SIZE INCH	RADIUS INCH
1/4 TO 1/2	1/32
9/16 TO 1	1/16
1-1/8 TO 1-1/2	1/8



NOMINAL SIZE (INCH)	ANGLE "W" (DEGREES)
1 AND UNDER	10
OVER 1	6

C = CLEARANCE OF WEDGE, INCHES.
D = NOMINAL DIAMETER OF BOLT.
R = MAXIMUM RADIUS, INCH.
T = THICKNESS OF WEDGE AT SHORT SIDE OF HOLE EQUALS 1/2 (D).
W = WEDGE ANGLE (DEGREES).

Figure 3 - Wedge tensile strength test details.

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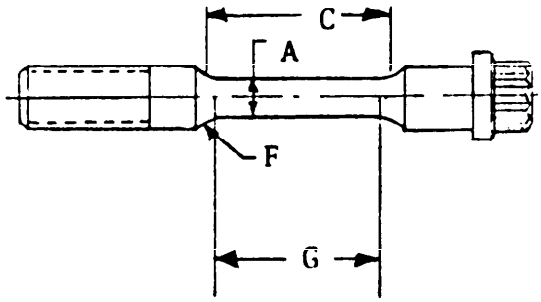


Figure 4 - Tension test specimen for screw with turned-down shank. (See FED-STD-151 for dimensions).

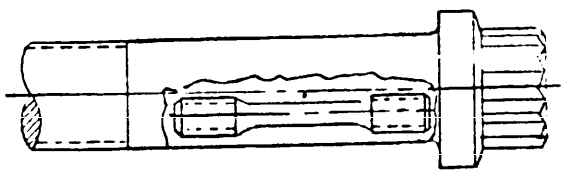
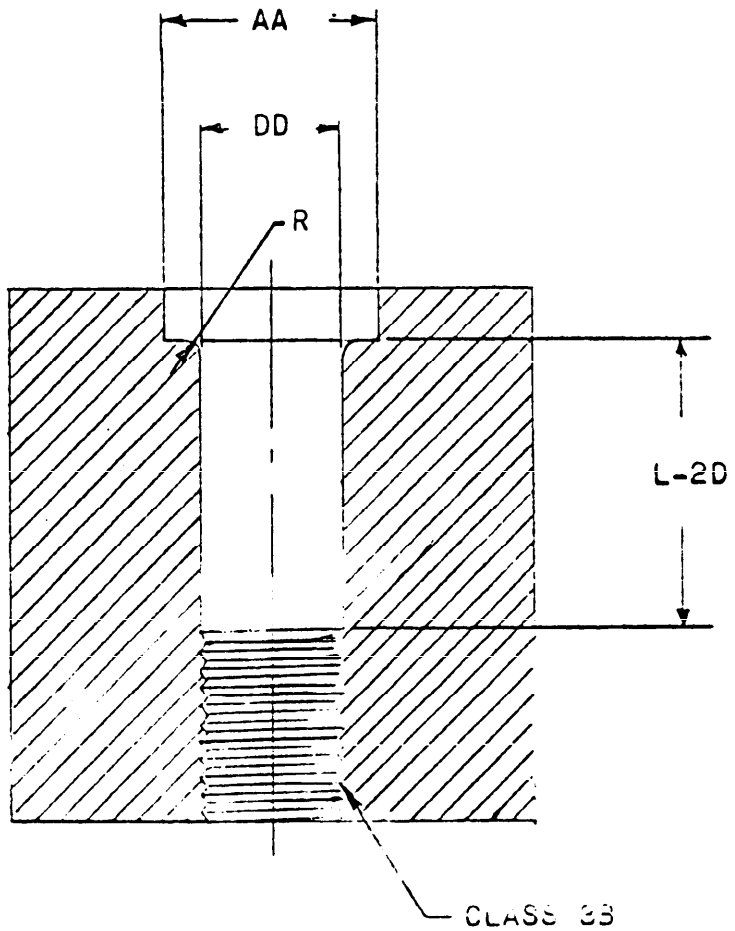


Figure 5 - Location of standard round tension test specimen when turned from large-size screw.

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The collar shall be concentric with the body within 1 percent of the maximum screw diameter, (2 percent total runout) or 0.010 inch total runout, whichever is greater.

Total runout between thread, body and collar shall be checked by assembling screws into a compound gage which is threaded at one end to the basic thread size (class 3B) counterbored at the other end to diameter AA listed on the specification sheets and whose center portion is equal to DD. Diameters AA and DD shall be concentric with the axis of the thread. The length of the center portion shall be equal to the screw length minus two screw diameters. The starting thread shall be chamfered and the junction between diameters AA and DD shall be rounded to a value equal to R maximum.

Figure 6 - Concentricity.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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